



Public Health ROI Analysis

Sacramento County | Fiscal Year FY 2024

Key Findings

Sacramento County's public health investment of \$21.2M is estimated to save 194.2 lives annually, generating \$2.6B in social value.

194.2

Lives Saved (Annual Estimate)

Based on statistical relationship between spending and mortality

125:1

Benefit-Cost Ratio

Dollars of benefit per dollar invested

\$2.6B

Social Value

Monetary value of lives saved using federal guidelines

\$109K

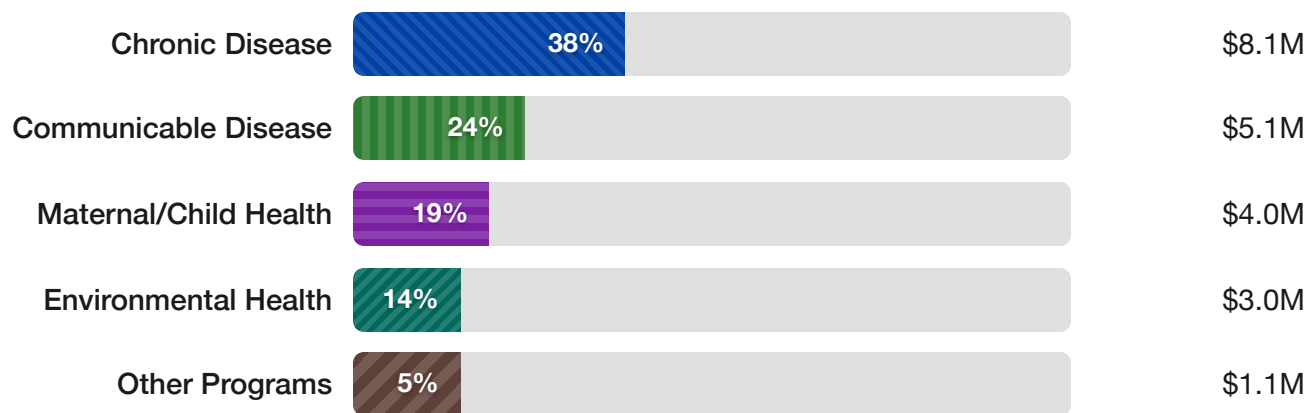
Cost Per Life Saved

Investment required to save one statistical life (vs. \$13.6M federal VSL)

Total Investment: \$21.2M (\$13.62 per capita)

Population Served: 1,556,683

Spending by Program Area



Program Area	Amount	Percentage
Chronic Disease Prevention	\$8.1M	38%
Communicable Disease Control	\$5.1M	24%
Maternal & Child Health	\$4.0M	19%
Environmental Health	\$3.0M	14%
Other Programs	\$1.1M	5%
Total	\$21.2M	100%

Methodology

This analysis estimates how public health spending affects mortality rates and translates that effect into lives saved and economic value.

The underlying research uses Lewbel IV estimation on California county panel data (2003-2023) to identify the causal effect of public health spending on mortality rates.

Key Finding: Each \$10 per capita in public health spending reduces mortality by 9.16 deaths per 100,000 population.

For context: if a county of 1 million people increases spending by \$10 per person, approximately 92 fewer deaths would occur annually.

Social value is calculated using the U.S. Department of Health and Human Services Value of Statistical Life (VSL): **\$13.6 million** (2024 dollars).

Data Sources

- Expenditure data: Comprehensive Annual Financial Report (CAFR)
- Population data: California Department of Finance
- Mortality coefficient: Lewbel IV estimation (Cholette, Patton & Zaragoza-Gonzalez, 2026)

Limitations

These estimates are based on historical county-level data. Actual results may vary based on program implementation. The coefficient represents an average effect across all public health activities; specific interventions may have higher or lower returns.

This report is designed for presentations to boards, budget committees, and other stakeholders. Full methodology is available in the attached working paper.

Full Citation:

Cholette, V., Patton, D., & Zaragoza-Gonzalez, M. (2026). The Causal Effect of Public Health Infrastructure Spending on Mortality: Evidence from California Counties. SSRN Working Paper.

Interested in deeper analysis? CAPHE is conducting ongoing research into ROI variation across program areas. Partner with us and receive more detailed, individualized findings for your county. Contact info@caphegroup.org to learn more.

California Association of Public Health Economists (CAPHE)

www.caphegroup.org | info@caphegroup.org

